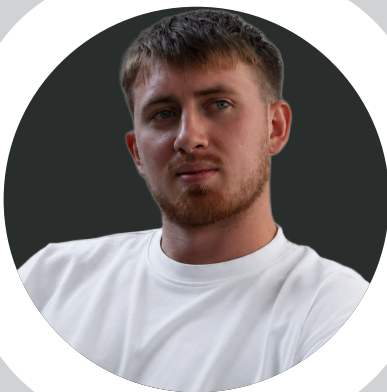




Jakub Cerovsky

class Robotics Engineer:

CONTACT

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- @ jacobcerovsky@gmail.com
- 🌐 JakubCerovsky
- 🌐 Jakub Cerovsky
- 🏠 Kongens Lyngby
- 📅 April 29, 2002

LANGUAGES

- Czech: Native
- English: Fluent (C1)
- Danish: Module 4 (out of 5)

SKILLS

Hard Skills

- Python
- TensorFlow
- PyTorch
- Computer Vision
- C++
- Git
- C#
- TypeScript
- AWS
- Docker

Soft Skills

- Eagerness to learn
- Curiosity
- Communication
- Team work
- Stakeholder Management
- Dependability

📄 Professional Profile

Software Engineer and MSc student in **Autonomous Systems** with hands-on experience in robotics, embedded systems, and software development. Currently working as a **Student Software Developer** at **the LEGO Group**, contributing to production software and collaborative development workflows. Background in **Python**, complemented by experience with C/C++, C#, TypeScript, and modern software engineering practices.

I particularly enjoy working at the intersection of software and physical systems, where code can be tested and improved through real robot behaviour. I am looking to develop further as a **Robotics Engineer** while bringing my existing software engineering experience into hands-on robotics development.

📁 Professional Experience

Student Software Engineer, The LEGO Group, Edge Platform

August 2023 – Present

Develop and maintain production software across **frontend**, **backend**, and **edge systems** using **TypeScript**, **React**, **AWS**, and **Docker**. Introduced **network-condition simulation** into end-to-end testing of physical edge devices, improving robustness under unreliable and constrained network conditions across global production sites. Automated parts of the **API development workflow**, simplifying endpoint integration. Collaborated within an engineering team using **Git**, **CI/CD**, code reviews, and automated testing.

Bachelor Thesis – Software Engineer, GLS Denmark, EV Charging Optimization

March 2025 – December 2025

Developed an **EV charging optimization platform** for GLS Denmark as part of a three-person engineering team. Built a distributed **microservice architecture** integrating electricity-price **APIs** and **MongoDB**, with a **single-page web application** for managing and visualizing optimized EV charging schedules.

🎓 Education

Master's degree, Autonomous Systems

DTU | Technical University of Denmark, Kongens Lyngby

February 2026 – Present

Bachelor's degree, Software Technology Engineering VIA University College, Horsens

August 2022 – January 2026

AP degree, Computer Science Zealand Academy, Roskilde

August 2021 – July 2022

Relevant Projects

Robotics Software Engineer, DTU, Robot Competition

Developed software for a **mobile robot** competing in a multi-challenge robotics track. Implemented **line-following and junction detection** in **Python** using an 8-element reflectance sensor and developed reusable robot functionality including servo control. Worked with **Raspberry Pi, MQTT, SSH, and Git**. The final robot placed **4th out of 15 teams**.

Robotics Safety Engineer, DTU, Safety Analysis of an Autonomous Agricultural Robot

Performed a **risk and functional-safety assessment** of an autonomous agricultural robot, covering mechanical, navigation, perception, control-system, and human-interaction hazards. Applied **ISO 12100, ISO 13849-1, and ISO 25119** principles to hazard identification, risk estimation, and risk reduction for autonomous operation.

Embedded Machine Learning Engineer, DTU, Embedded Hand-Sign Recognition

Developed an **embedded computer-vision and machine-learning system** for real-time hand-sign recognition on a resource-constrained **XIAO ESP32-S3 Sense** microcontroller. Built a **Python/TensorFlow** training pipeline and optimized and quantized a **Convolutional Neural Network** for embedded deployment.

Software Engineer, DTU, High-Performance Computing & GPU Optimization

Profiled and optimized a **Python Jacobi solver** for large-scale thermal simulations using CPU and GPU parallelization. Implemented **multiprocessing, Numba JIT, CUDA, and CuPy** solutions and developed a custom CUDA kernel with optimized memory access, achieving a **13.96× speed-up** over the reference implementation.

About me

I am **24** years old, born in the **Czech Republic**. I live in **Kongens Lyngby**, where I also attend the Technical University of Denmark.

My biggest passion outside of my career is being a hybrid athlete. After completing **IRONMAN** Copenhagen, two **IRONMAN 70.3** races, and several shorter triathlons, I continue to look for new physical challenges alongside my professional ones.

As an extracurricular activity, I like to attend **hackathons** and networking events such as **social runs**. Furthermore, I like to boost my creative side by playing on a **Minecraft** server in collaboration with my friends.

I very much enjoy a good beer with a **football** match or **BBQ** in the background and any kind of delicious **food**!

ACHIEVEMENTS

IBM Cloud Computing
Cambridge C1 English
IRONMAN Finisher

HOBBIES

Football
Triathlon
Puzzles
Travelling
Social Runs
Cooking & Eating
Motorbike